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# Umberto Domenico Ciccia

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## Experience

**Amazon Web Services (AWS)**, Dublin, Ireland, Nov 2025 – present
**System Development Engineer**

- Led the refactoring of 184 Route 53 infrastructure pipelines, increasing pipeline freshness from 80% to 98% in eusc-de-east-1.
- Unblocked a strategic customer launch under C-level executive escalation by delivering DNSSEC support for Route 53 in AWS European Sovereign Cloud 3 weeks ahead of schedule, enabling migration without changes to the customer's existing infrastructure.
- Redesigned DNS infrastructure, eliminating a recurring misconfiguration and reducing SEV-2 alerts from 84 to 0 over six months across all AWS partitions.
- Maintained reliability of Route 53, AWS WAF, AWS Shield services in sovereign regions, achieving 99.99% availability through incident response, on-call rotations and RCA-driven improvements.
- Delivered the Route 53 CIDR routing feature end-to-end in the European Sovereign Cloud 4 months ahead of schedule, implementing infrastructure, APIs, and console support for customer-configurable IP-based DNS routing.

**NTT Data**, Cosenza, Italy, Jan 2025 – Oct 2025
**Cloud Engineer**

- Built GCP data ingestion pipelines processing 10k+ daily records, improving data availability and release reliability using Dataflow, Compute Engine, and Cloud Storage for Coop Italia
- Designed and deployed a multi-cloud infrastructure spanning AWS and on-premises using Terraform, Kubernetes, EKS, Aurora, DMS, S3, and ECR, reducing deployment failures by 70% and improving deployment consistency for ItaliaOnline
- Configured on-premises Jenkins to manage CI/CD pipelines for new containerized application versions deployed to EKS, fully automating releases and deployment (100%) for ItaliaOnline.
- Reduced infrastructure costs by 30% by building an AI-powered Quarkus agent with the Vertex AI SDK to estimate costs and recommend optimized multi-cloud deployments for user-defined architectures on the NTT KUMO platform.

**Vision One**, Recanati, Italy, July 2024 – Dec 2024
**Software Engineer**

- Developed scheduling system in .NET/Blazor, improving task allocation efficiency by 80%.
- Refactored legacy codebase, reducing unused code by 60% and improving maintainability.
- Introduced automated testing (xUnit), increasing test coverage to 80%.

**Caliò Informatica**, Cosenza, Italy, Jan 2024 – July 2024
**Software Engineer**

- Developed workflow-building capabilities for AIDA, an OpenAI-powered automation platform, enabling non-technical users to create AI workflows through a visual drag-and-drop interface inspired by n8n.
- Optimized OpenAI API consumption through request redesign and prompt engineering strategies, reducing operational costs by 80% while improving response performance by 60%.

## Skills

**Programming Languages:** Java, TypeScript, JavaScript, Python, C#, SQL, Bash, PowerShell, C++
**Frameworks:** Spring Boot, Quarkus, .NET, FastAPI, Django, React, Angular, Blazor
**Cloud & DevOps:** AWS, Azure, GCP, Kubernetes, Docker, Helm, Terraform, Pulumi, AWS CDK, Ansible, Argo CD, Jenkins
**Databases:** PostgreSQL, SQL Server, MongoDB, Redis, DynamoDB
**Tools & Technologies:** Git, Linux, NGINX, Istio, REST APIs, OpenAPI, gRPC, Prometheus, Grafana, Testcontainers, JUnit, SonarQube
**Languages:** Italian (Native), English (Professional)

## Certifications

- AWS Certified Solutions Architect – Associate
- Certified Kubernetes Administrator (CKA)
- Google Associate Cloud Engineer
- HashiCorp Terraform Associate

## Education

**Università della Calabria**, M.Sc. in Artificial Intelligence and Machine Learning – GPA: 4.0/4.0

- Relevant coursework: Artificial Intelligence, Machine Learning, Computer Vision, Distributed System, Knowledge representation and reasoning, Statistical Learning

**Università della Calabria**, B.Sc. in Computer Engineering – GPA: 4.0/4.0

- Relevant coursework: Operating Systems, Computer Networks, Database Systems, Object Oriented Programming, Software Engineer

**LeadTheFuture**, Mentee

- Chosen as one of the few Italian students (acceptance rate <20%) for LeadTheFuture, a prestigious non-profit mentorship program for high-achieving STEM students.